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Predict Solar Panel Power Production Using Weather Forecasts

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***Abstract*—Solar energy is one of the widely used renewable energy in homes and industries and could replace fossil fuels. Frequently changing weather conditions are the main factor that affects solar power generation. Sri Lanka is a tropical country that is affected by unpredictable weather conditions. It is situated very near to the equator and receives more sunlight. Many solar panel users in Sri Lanka generate electricity for ordinary households and industrial purposes. Solar panel users must rely on fossil fuel energy when solar power generation decreases. It increases the load on the main grid.**

This research focuses on how weather conditions affect solar panel productivity and how solar panel users can use renewable energy in an efficient way using accurate weather prediction systems. Weather data and solar panel energy production data were collected and analyzed for this research. A Google form was distributed to the solar panel users to collect data about the weather prediction systems that are used to estimate solar panel power production. First, the correlation between weather conditions and solar energy production was analyzed. A Neural Network(NN) and the SARIMA algorithm are used to create the weather prediction system and compare the accuracy. Then, the weather prediction system is used to identify the increment and decrement in solar panel power production. This research aims to transform the weather prediction system into a solar power prediction system and use solar power efficiently to manage the load on the main grid.

***Index Terms*—Weather Prediction, Solar Energy Prediction, Weather Forecast, Data Analytic.**

I. Introduction

The climate crisis is one of the important issues in this world. Climate change affects many people's daily lives and costs many lives. [1] In 2018, the Intergovernmental Panel on Climate Change (IPCC) became aware that fossil fuel is the cause of 89% global CO_2 gas emissions. This is one of the significant reasons for the climate crisis. [2] According to the International Energy Agency (IEA), 60.19% of electricity is generated in the world by burning fossil fuels. This usage in electricity generation releases a large amount of CO_2 and various greenhouse gases like Sulfur dioxide (SO_2) and nitrous

climate crisis. Nuclear energy and renewable energy sources are the solutions for substituting fossil fuel usage in electricity generation. [2] [3] 10.12% of electricity is produced in 2020 using nuclear energy. Nuclear energy causes environmental problems and produces 400,000 tons of nuclear waste per year worldwide, according to the World Nuclear Association. It is difficult and expensive to dispose of this nuclear waste. If this nuclear waste is released into the environment, then it is difficult to remove those radioactive particles from the environment. Natural disasters such as floods, tsunamis, and earthquakes could easily damage nuclear plants and cause the release of radioactive particles into the atmosphere. [4] Those radioactive particles take 1,000–10,000 years to decay. So, this nuclear plant's energy could not be the best substitute for fossil fuels. There are various renewable energy sources for example wind, solar, hydro, and geothermal energy, which can be used for electricity generation. Renewable energy is environmentally friendly and renewable. [3] In 2020, around 28.97% of the energy used in the world was produced from renewable sources. [3] In total renewable energy generation, 60.08% of energy is generated from hydropower. Only 10.3% of energy is generated using solar power and can increase to a certain level.

Countries close to the equator get more efficient solar energy than countries far from the equator. It happens because of the inclination of the earth's axis. [5] Sri Lanka is close to the equator and it is the 25th largest island, which has 65,612 km2 of land. Sri Lanka could use solar energy for electricity generation rather than fossil fuels. Sri Lanka is a tropical country that gets more rainfall and has very quickly changing weather conditions.

Sri Lanka is divided into the wet zone and dry zone according to the weather pattern and rainfall. The Western, Southwestern, and Central provinces fall into the wet zone, which receives an average of 2500 millimeters or more of rainfall annually. The southwestern monsoon brings more rainfall to the wet zone. The Northern, Northwest, Southeast, and East country parts come into the dry zone, which annually gets between 600 and 1900 mm of rainfall [5]. The dry zone gets rain mostly from the northeast monsoon. The dry zone is better for solar energy generation because bad weather does not affect solar energy generation as much as it does in the wet zone. The weather conditions of the wet zone affect solar power generation.

Western Province is located in the wet zone and is one of the major cities and commercial capitals of Sri Lanka. Many solar panel users in the Western province have fixed solar panels on their home rooftops, and some industries also use solar panels to produce electricity. These solar panel users

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could not produce sufficient electricity during bad weather conditions or cloudy times. So, they rely on the main grid for electricity usage. When the solar panel users rely on the main grid load, the main grid will increase. It is mandatory to use the solar panel power generation prediction system to balance the load on the main grid. The fuel crisis halted the production of fossil fuel electricity in countries such as Sri Lanka and Europe. Even the generators don't have the fuel to produce electricity. Even hospitals can't operate their medical electric equipment because of the power crisis in Sri Lanka in 2022. Solar panel systems can provide support at these times. The solar panel system can help to produce electricity for countries that have a fuel crisis. Currently, solar panel users directly use weather prediction systems to predict solar power generation. They directly depend on mobile and web weather prediction systems and the weather forecast news. Experienced solar panel users could be able to predict the solar panel's power generation using weather prediction systems. Using weather prediction systems, novice solar panel users cannot accurately predict solar panel power production. This research analysis shows how solar panel power generation is affected by weather conditions. The efficiency of the weather prediction system is analyzed to predict solar power generation. The Google form was distributed to solar panel users in Sri Lanka to confirm the use of weather predictions.

The specific objectives of this research paper are as follows:

- Analyze the correlation between atmospheric conditions and energy produced using solar.
- Analyze the efficiency of weather prediction systems that can help to predict solar power production.

II. Related Research

Renewable energy is uncontrollable and depends on various atmospheric conditions [6] [7] [8] [9]. So users should depend on the grid. Solar power generation's variable output creates uncertainty on the grid. It affects the operation of the electricity grid. The grid should continuously monitor the electricity demand. According to it, generators should be dispatched to fulfill the electricity demand as it increases and falls. Some other prediction models could improve by adding more weather phenomena as inputs [9] [6].

Climate change creates the need for weather forecasts. The data mining model was used for weather forecasts. The data mining model was used and built on the data that included past weather records, day-to-day rainfall, maximum and minimum temperatures, air temperature, relative humidity, wind speed, soil moisture, soil temperature, and other factors that affect precipitation. First, A. Omary, A. Wedyan, A. Zghoul, A. Banihani, and I. Alsmadi [10] applied data preprocessing techniques to prepare the data. The main goal was to introduce a weather forecast model based on the High-Resolution Limited Area Model (HIRLAM) which is a numerical weather forecast model for the short-range weather forecast and ALADIN models, especially for precipitation prediction. A software robot program that collected all weather-related information from worldwide websites was built. Later, the Jordan precipitation was predicted using that information via data mining techniques and AI algorithms.

In this research [11] M. Biswas, T. Dhoom, and S. Barua used the Apache Hadoop framework and Map-Reduce framework and predicted the rain with the help of the Naïve Bayes Algorithm. Precipitation and humidity are mainly used for predicting rain. They [11] mainly focused on analyzing big data using Split, Load, and Store commands. They [11] plotted the graph for the precipitation and the humidity. They then analyzed the relationship and plotted the graph of rainfall. In the next version, they [11] used the Naïve Bayes algorithm in the Hadoop Framework to successfully predict the rainfall. This system took weather data as big data input and forecasted rainfall with min, max, and mean rainfall efficiently.

This research [11] used the Naive Bayes classification technique and Chi-Square strategy for weather forecasting. Using current weather conditions, the proposed system forecasts future weather data.In the initial stage, the data was collected and preprocessed to contain the weather parameters like Outlook, temperature, humidity, and wind. Then M. Biswas, T. Dhoom, and S. Barua [11] store it in the MySQL database system. Naïve Bayes helped to forecast the weather into 2 classes: Good or Bad. The study [11] proposed a system that could be implemented using Java and store the data using MySQL Server. Finally, the recall and precision of their predictions were analyzed to validate the model. This research [11] concluded that more attributes of weather conditions needed to be considered for accurate prediction.

A precise weather prediction system could help to reduce the losses from natural disasters like tropical cyclones and floods. First, the issues were analyzed in the weather forecasting. Next, P. C. Reddy and A. S. Babu [12] analyzed the different methodologies and machine learning algorithm used to predict the weather forecast based on author, assumption, dataset, time duration of investigation, methodology, input parameters, and recommendations. The main goal of this analysis is to find early weather prediction models. This research [12] observed that Map Reducing Algorithm and linear regression methods provided better results than other various methods. The neural network performs well for yearly basis prediction, and FFNN produces good results on a monthly and weekly basis. The highest accuracy of 93% is produced using FCM. The outliers reduced the performance of the SVM.

This research [13] focused on predicting the weather data using SVM and Linear Regression with Hadoop. The weather data from 2005 to 2020 is collected by the INDIA METEOROLOGICAL DEPARTMENT (IMD) for this research. Linear regression and SVM were used with Big Data MapReduce to forecast the weather conditions. Sunny, cloudy, rainy, and Humid atmosphere conditions for Delhi, India using their research. The proposed model was better than the proficient ones over longer timeframes. For the future scope, Shubham Madan, Praveen Kumar, S. Rawat, and T. Choudhury [13] planned to use Apache spark for concurrent prediction of weather.

In this research, [14] Naveen L. and Mohan H.S. used Artificial Neural Networks (NN) and SVM. RBF-NN, RBF-GA and RBF-HPSOGA, Gaussian SVM, and Wavelet SVM models are evaluated in this study [14] to forecast the weather. The wavelet-based SVM and RBF NN utilizing HPSOGA gave remarkable execution productivity and had low RMSE among other models.

This study [15] mainly focused on artificial neural networks to forecast the weather. They [15] input wind, temperature, and rainfall. ZhanJie Wang and A. B. M. Mazharul Mujib successfully predicted the temperature and windspeed using the ANN in terms of the month and year. This research [15] explains how the ANN could be used to predict the weather. The result was compared with the last five-year weather data collected by the Dalian Meteorological Bureau. This data mining approach via prediction helps to solve the problems in the wind forecast for wind farms.

In this study, [16] the weather forecasted by the Artificial Neural network (ANN) because of its nonlinear characteristic. The data for this study was gathered and preprocessed from various websites. The model was created using ANN, trained, tested, and validated. The Back-Propagation approach was used the reduce the error. The sigmoid function was used in the hidden layer. The model predicted the minimum and maximum temperature of the day, relative humidity, and rainfall. The Android system was created, and it has 3 phases: the registration phase, the login phase, and the notification phase. The ANN with the back-propagation algorithm model was validated using the MSE. The model had a lower MSE and high accuracy than the other statistical models. The ANN accepted the complex parameters as input and generated the intelligent pattern.

The goal of this research [17] is to create a hybrid model that could be created using statistical (ARIMA model) and ANN techniques to perform better in predicting the weather. ANFIS model had better performance than the ARIMA model for Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R2). T2 FLS and FCM & T2, ANFIS and IT2TSK, GRANN_ARIMA models were also evaluated in this research. The hybrid method helps to reduce the error in prediction. Most complex models were used to gain accuracy, but the hybrid model can be used to decrease the execution time of these complex models.

This study [18] focused on the deep learning-based weather prediction system created with the Python Keras library, TensorFlow framework, and Pandas library. The data collected from NOAA National Climatic Data Centre (NCDC) included local temperature, precipitation, wind, station ID, station location, date, and other needed data. The accuracy of the prediction increases with more data (Hypothesis I), and more recent data will give more value to the prediction (Hypothesis II) were the hypotheses of the researchers. J. Booz, W. Yu, G. Xu, D. Griffith, and N. Golmie used a sliding window matrix-based mechanism with deep learning supervised training in the big data environment. The model was evaluated using the MSE. Hypothesis I was validated in the study, but Hypothesis II is incorrect. The most latest data was not significant impact to the final result. The study [18] was carried out for other locations in the US. From that, it [18] concluded that increasing training data decreased the accuracy.

This research [19] discussed deep learning in the weather forecast. CNN for observation, ConvLSTM for nowcasting, NWP model for pure AI forecasting, Hybrid AI NWP forecasting, Postprocessing of NWP Output, Multimodal Combination, Downscaling, Warning for High-Impact Weather, and CNN for Decadal Climate prediction discussed in this study [19]. Advantages of the Deep Learning Techniques to Traditional NWP methods are mentioned in this study [19]. The research [19] concluded that the DL model could be more efficient for human development effort, computing power, and high forecast quality.

This research goal [20] was to create a web app to get the weather forecast of any city using the JavaScript framework AngularJS. The AngularJS component usage was described clearly in this study. They used the MV* model for efficiency, reuse, clean code, and more manageability. This application can show the weather forecast for two, five, and seven days for any city using the data (wind speed, humidity, pressure, latitude, longitude, and so on) from the API.

The weather data was collected for Fortiss, Munich, and Meteoblue from 2013 to 2016. The data collected was analyzed for the research [6]. The objective is to use different machine learning techniques to generate models that can forecast energy generation for Photovoltaic cells. Support Vector Regression (SVR) and Random Forest (RF) algorithms were used in this research [6]. Then researchers, A. Bajpai and M. Duchon [6] evaluated the performance of the models by Root Mean Square Error (RMSE) and R squared values on the test set. The hybrid model had the best accuracy. They [6] combine clustering, classification, and regression techniques and used the K-means algorithm for clustering. The 2 clusters were labeled "cluster 1" and "cluster 2". Different regression models were used on different clusters to predict them. The hybrid model had fewer errors. Forecasting accuracy depended on the weather forecast accuracy, distance from the weather station, and model performance.

The research [7] assumed that the solar intensity is proportional to solar panel-generated energy. First, the historical weather data and historical forecast data were gathered for this research [7]. The researchers studied how the weather affects the solar intensity level. After that, machine learning models were trained to predict solar intensity. Every model's accuracy was checked using the Root Mean Squared Error (RMS-Error) between the forecasted and measured solar intensity. N. Sharma, P. Sharma, D. Irwin, and P. Shenoy [7] used Support Vector Machines, RBF kernel and Linear Least Squares Regression to predict the solar intensity with better performance. The redundant information was removed using Principal Component Analysis (PCA) to increase accuracy. Finally, PPF, Cloudy-FPF, and SVM-RBF Kernel were compared with reduced features. The SVM with RBF kernel and linear least-squares outperformed past-predicts-future models based on weather conditions.

The research [8] goal is to create a solar power generation forecast using the Artificial Neural Network(ANN). The weather data was gathered from the Global Energy Forecasting Competition - 2014 (GEFCOM-2014). Mohammad first prepared and analyzed the data for the study. The variable that has more impact was identified for solar power generation by finding the root mean square error (RMSE). The model was created using ANN and trained using the historical data except for the last month. Last month's data was used for validation. The model performance was evaluated by plots and graphs, RMSE, the correlation coefficient(R) between the actual and predicted result, and a comprehensive comparison with various models. The ANN model predicts better than other models. Forecasting with an hourly step size improves performance more than recursive neural networks. The model prediction is better in the clear sky than in cloudy hours. More accurate and additional weather data will produce more accurate predictions.

The historical weather data from 2003 to 2013 was gathered for the research [9] the Georgia Automated Environmental Monitoring Network (GAEMN) and hourly forecasted weather data from the National Oceanic and Atmospheric Administration (NOAA). 18 input weather phenomena were used in various machine learning models and predicted the solar radiation 1 hour in advance. By increasing the inputs and adding the surrounding area weather forecasts, the improvement was identified in machine learning models rather than input historical data alone. One-hour predictions were more efficient and observed relatively slight weather changes than 24-hour predictions. In this study, Sam used Linear Regression, Neural Network, M5P model tree (unpruned and pruned), Random Tree, REP Tree (unpruned and pruned), Bagging M5P (unpruned and pruned), and Random Forest for 1-hour and 24-hour prediction. In both cases, Random Forest resulted in low MAE (watts/m2).

This research [21] focused on predicting solar panel power output using Elman Neural Networks (ENN) and Feed Forward Neural Networks (FFNN) which use the Levenberg-Marquardt activation function. The input included the PV power output (W), module temperature (°C), ambient temperature (°C), and solar radiation (W/m2) from March 22 until April 2, 2019. They [21] used 3 input vectors based on 3 cases with different variations. After the models were trained, K. b. Hanifulkhair, A. Priyadi, V. Lystianingrum, and R. Delfianti evaluated the model based on the RMSE. The researchers plotted the graph using predicted values and actual values for the 3 cases and evaluated the graphs to find a suitable model. This research [21] concluded that the FFNN performs better than the ENN based on the model that had a small MSE.

This study [22] proposed a smart solar home system that had battery utilization with the forecast methodology to optimize the battery power usage in daily and seasonal variations. A. Manur, M. Marathe, A. Manur, A. Ramachandra, S. Subbarao, and G. Venkataramana [22] identified the energy surplus, energy deficit, and loss of communication issues in SHS that can be solved using the Solar Power Forecast. The research used fifty eight days of weather and solar power production data that had temperature, sunrise time, sunset time, cloud coverage, battery voltage and respective timestamp. The dataset was preprocessed then used to train the Long Short Term Memory (LSTM). The hyperparameters were fine-tuned to reduce the MSE. The model was evaluated using RMSE. For the whole forecast set of 11 days it provided an RMSE of 0.11. This prediction model could help with grid predictive maintenance and demand optimization. This could be helped to omit the system failures and the reliability of the grid system increased.

This research [23] used the NN with the backpropagation algorithm to forecast solar power output. MATLAB was used to analyze the data that contained the climatic conditions, humidity, temperature, and humidity. The backpropagation algorithm helped to correct the weight and make the prediction close to the actual value. Their [23] model shows that the power output reaches peak at a high temperature. The cloud presence, dust, and temperature can be brought together under a single network to predict solar power more accurately.

The article [24] analyzed the scientific literature and publications on photovoltaic solar power management from 2000–2019 using bibliometric method. First, T.M. David, P.M.Silva Rocha Rizol, M. A. Guerreir oMachado, and G.P. Buccieri [24] searched in the Scopus database to gather a set of data for all published articles using the following keywords: ”solar”, ”energy”, ”management” and ”photovoltaic”. The terms, publications, authors, and country were analyzed using Scimat and VoSviewer software. Research [24] identified that the US, China and India have a significant impact in this field. The keyword study and the content study were performed in this research using the VoSviewer software. The researchers [24] found new trends in photovoltaic solar energy management to bring study opportunities. Previous publications on solar energy will mainly support the technical aspects and socioeconomic aspects.

The research [25] focuses on reducing the delay in transferring the weather data via cables and other physical mediums for the weather forecast. The researchers focused on reducing the cost of implementation and increasing the speed of the weather forecast system using various sensors that can communicate using wireless communication.

This related research paper concluded that weather data is very important in predicting solar panel power production. The forecast weather data is the main ingredient for predicting solar panel power generation.

III. Methodology

A. Data Collection

The weather data used in this research was collected from the Katunayake/Bandaranaike airport. The weather data consists of temperature, atmospheric pressure, relative humidity, wind direction, wind speed, total cloud cover, visibility, and precipitation. The collected weather data didn't consist of solar radiation measurement-related measurements. The solar panel power generation data was collected from the CEB(Ceylon Electricity Board). The weather data and solar panel power generation data were used to find the relationship and validate the hypothesis between the weather conditions and solar panel power generation. This data is used to analyze the feasibility of how the weather prediction system helps solar panel users predict solar panel power generation. Other than this data, the Google form was prepared and distributed among the solar panel users in Sri Lanka. The Google form was distributed to the solar panel users. Figure. 1 shows the people who filled out the Google form and their experience with solar panel usage.

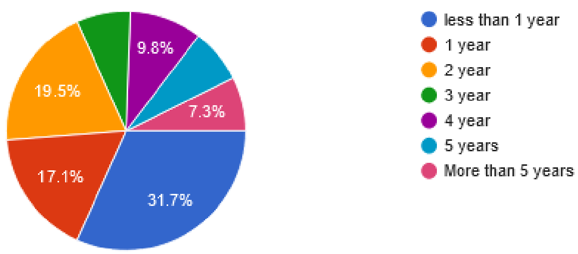

Fig. 1: Participant in Google Form Survey: Years of Experience with Solar Panel Usage

B. Data Analysis

First, the collected data was analyzed carefully. In the solar panel power generation data, April data was missed because of the pandemic situation in Sri Lanka. The whole country went into full lockdown. According to the situation, the electricity board didn't calculate the value for April. This situation created the null value in the data set. At the same time, the 2020 solar panel meter value wasn't collected regularly. It created unnecessary values in the data values. For this reason, the 2020 solar panel dataset was removed from the overall dataset. The solar panel power generation dataset for 2019 was only considered for this research. Using data visualization techniques, the solar panel power generation chart was prepared. Please refer to Figure. 2 to understand how solar panel power production differed according to the month. The change in the number of solar panel users and the number of connected solar panels from January 2019 to December 2019 is considered as not changed.

At the same time, the weather conditions of 2019 were analyzed to get insights that could help to identify the relationship between solar panel power production and weather conditions. The solar irradiance measurement was missing in the collected weather data. The multiple weather condition charts and the date charts were created using suitable data visualization techniques. Then the weather conditions charts were compared with the solar panel power production chart (Fig. 2). According to the analysis, precipitation affects solar panel production directly. The low solar power was produced when the high precipitation and low temperature were measured by the weather monitoring system.

In the atmosphere, the water vapor is converted to liquid or frozen form and then falls back considered as precipitation such as rain and snow. [26] According to Sri Lanka, rain is only considered precipitation. Normally, the clouds are white. When the cloud thickness increases by observing more water vapor and droplets, It becomes gray instead of white. These thicker clouds can deflect the sunlight in various directions or absorb sunlight's energy and then re-radiate to different directions but not towards the land. This effect reduces the sunlight and causes poor solar power production. The clouds would have more water droplets, be darker, and cause high precipitation in the weather monitoring system. This is the hypothesis that is being considered for this research.

The temperature affected the solar panel power production by a very small amount. Low-temperature changes were monitored in May when low solar power was produced. Solar panel power production efficiency can be affected by a few other variables other than weather conditions like solar radiation and temperature. The factors are

- Solar panel reflection property.
- Shading from nearby trees or other buildings.
- Excessive dirt, dust, and pollution.
- Solar panels are facing direction.

This research only considered the weather conditions that affect solar panel power production. The dirt, dust, and pollution could reduce solar panel power production. The rain helps to remove the dirt and dust from the solar panels. This creates a sudden increase in solar panel power production after the precipitation has been monitored. In future studies, the effect of dirt, dust, and pollution could be considered in predicting solar panel power production predictions.

The above analysis confirms that weather conditions like

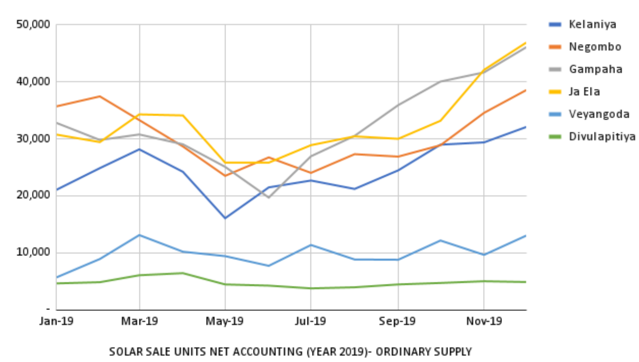

Fig. 2: Overall Solar Panel Power Production for 2019 (Western Province - Sri Lanka)

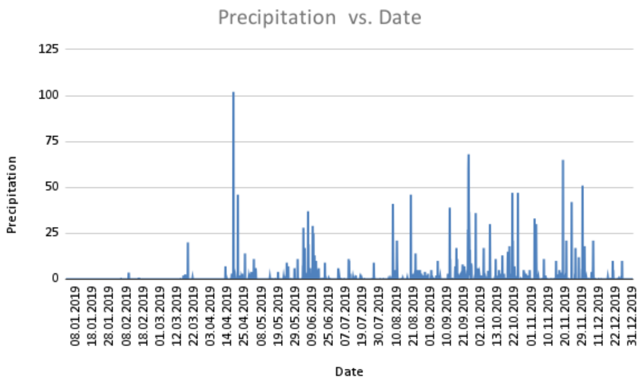

Fig. 3: Precipitation of 2019 (Katunayake/Bandaranaike Airport Weather Data)

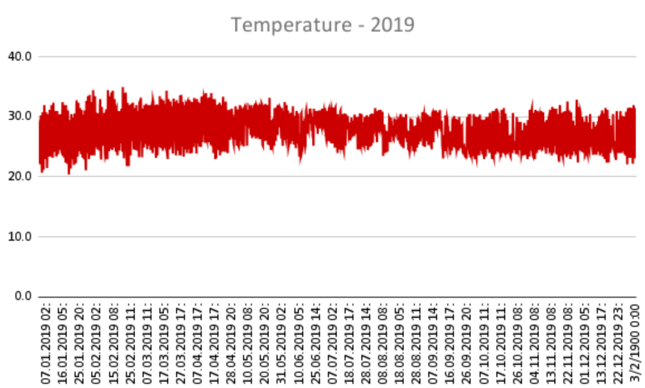

Fig. 4: Temperature of 2019 (Katunayake/Bandaranaike Airport Weather Data)

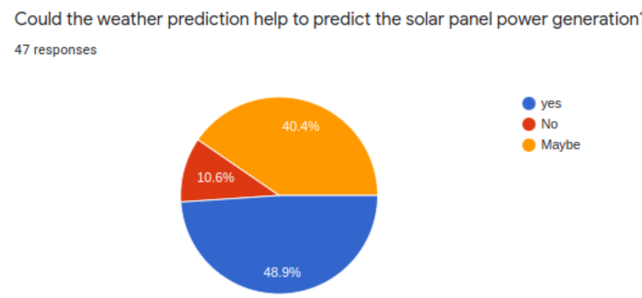

Fig. 5: Google Form Survey Result Validate the Weather Prediction System Usage

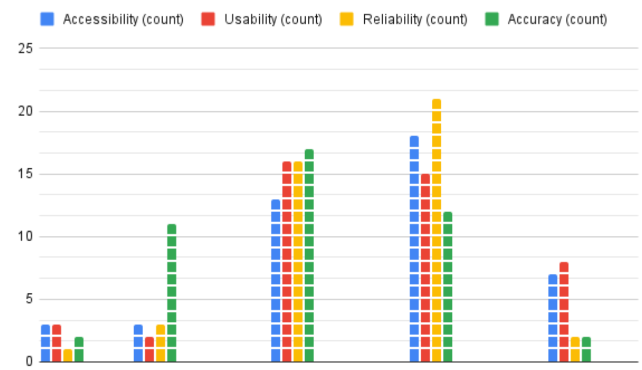

Fig. 6: Google Form Survey Result About the Weather Prediction System Quality Attributes

precipitation and temperature affect solar panel power production. The available weather prediction systems like AccuWeather can predict the precipitation and the temperature that affect solar panel power generation for 10 days [27]. The collected Google form data from experienced solar users validated and verified that the weather prediction systems could be used to predict solar panel power production. The collected Google form results are shown in Figure 5.

The quality attributes like accessibility, usability, reliability, and accuracy of the weather prediction system are evaluated to identify the things that need to be improved in the weather prediction system. The outcome is shown below in Figure 6. The weather prediction systems that provide the predicted weather data for a month won't have good accuracy. This results in low satisfaction with the accuracy of the weather prediction system for the Google form.

Long Short-Term Memory (LSTM) neural network and the SARIMA algorithm are used to create the models that can predict the temperature and precipitation for the solar panel power prediction. The predicted temperature and actual temperature are used to calculate the model RMSE values. The models are evaluated using the calculated RMSE values. The model that was created using SARIMA has 0.5271614479099459 RMSE value, and the LSTM model has 0.5575150711849083 RMSE value. According to the RMSE values, the SARIMA model has higher accuracy than the LSTM model. Figures 3-4 and Figures 7-8 show the monthly actual temperature and the predicted temperature of the LSTM model and the SARIMA model. The predicted temperature and precipitation can be used to forecast the increase and decrease in solar panel power production.

IV. Discussion

This research shows that temperature and precipitation have a direct impact on solar panel power production. Creating a highly accurate weather prediction system that can predict the temperature and precipitation will help to predict solar panel

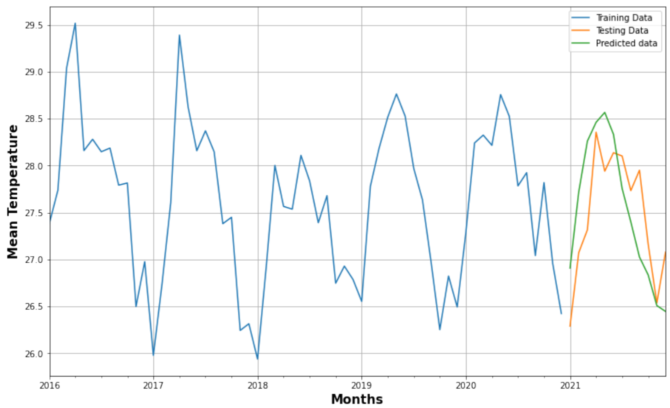

Fig. 7: The predicted Temperature using LSTM

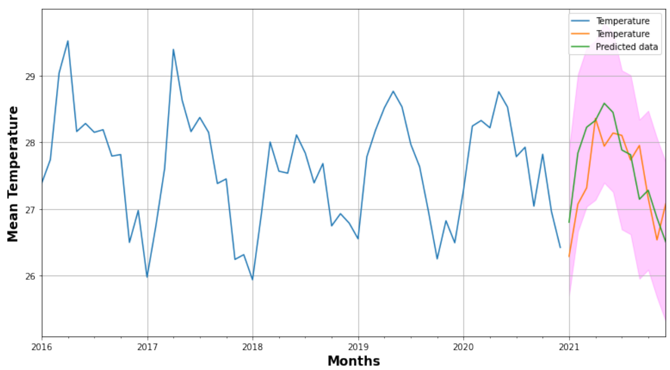

Fig. 8: The predicted Temperature using SARIMA

power production changes. The solar panel power production system that is created on top of these weather prediction systems can predict the temperature and precipitation. This system can be integrated with the existing systems like the CEB mobile application and website and can help to notify the sudden decrease and increase in solar panel power production to the solar panel users and the interested parties in solar panel power production, like CEB and the industries that depend on solar panel power production.

This particular system is easy to implement with the already existing weather forecast systems that provide the predicted temperature and precipitation for upcoming days. Apart from that, the necessary weather conditions data can be collected from the weather forecast websites using web scrapping technologies or directly from the weather forecast APIs. The accuracy of this system will depend on the accuracy of the collected weather data.

The notification will assist solar panel users in making efficient use of electricity during periods of low solar power production. Solar panel users can switch off high-energy-consuming appliances like refrigerators, washing machines, and ovens to save battery power. This helps to reduce the chance of being dependent on the main grid. They can check and confirm that the lithium batteries store electricity at their maximum capacity before the low power production time. Electricity loss can last a few hours to several days in a disaster situation like a storm and heavy rain. The current weather prediction system can be able to predict storms and heavy rainfall up to 5 days in advance with an accuracy of 90% [28]. This helps to store the electricity in the lithium batteries before the disaster happens. The stored electricity will help to survive a disaster. This is very helpful to hospitals in saving patients' lives.

Electricity producers like CEB can use solar panel power production predictions to assume the demand or load on an electrical main grid. This prediction system can help to implement profitable solar panel energy production in a wide range of energy production.

V. Conclusion

This research concludes that temperature and precipitation affect solar panel power production. The importance and applicability of converting the weather prediction system into the solar panel power prediction system were analyzed. The solar panel power prediction system has a significant impact on personal and national development was discussed in this research paper.

The future research gap is identified to include more weather conditions like cloud presence and solar panel location factors like dust, dirt, and pollution. The change in the solar panel users can be considered and improve the validation. The weather forecast accuracy needs to improve. There are various obvious ways to develop upon the analysis proposed in this research.

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